

INSTITUTIONAL MARKET RISK COMPLIANCE AUDIT

Basel III Regulatory Backtesting & Value-at-Risk Audit • Generated: 2026-08-06 14:46:51 UTC

1. EXECUTIVE PARAMETERS & PORTFOLIO TARGET

Target Asset Symbol	AAPL	Basel III Traffic Zone	GREEN ZONE
Portfolio Capital Value	\$1,000,000.00	Kupiec POF p-Value	0.9071
VaR Confidence Level	95.0%	EVT Volatility Cap	69.26%

2. QUANTITATIVE RISK METRICS (1-DAY HORIZON)

Metric Name	Mathematical Specification	Value (%)	Capital Amount (\$)
Predicted Volatility (5D)	LightGBM + EVT Capping	45.19%	-
Value-at-Risk (1D VaR)	Filtered Historical Sim (FHS)	5.09%	\$50,884.71
Expected Shortfall (1D ES)	Artzner Coherent Tail Loss	7.64%	\$76,424.17

3. BASEL III KUPIEC POF BACKTESTING MATRIX

Backtest Parameter	Empirical Result	Regulatory Threshold	Compliance Status
Total Historical Sample	471 Trading Days	>= 250 Days (Basel III)	PASS
Out-of-Sample Violations	23 Breaches (4.88%)	5.00% Expected Rate	PASS
Kupiec LR POF Statistic	0.0136	< 3.841 (Chi-Square df=1)	PASS
Kupiec Test p-Value	0.9071	> 0.05 (Non-Rejection)	PASS (Green Zone)

OFFICIAL REGULATORY COMPLIANCE CERTIFICATION:

This document certifies that the quantitative market risk models, volatility peramalan equations, and Filtered Historical Simulation (FHS) Value-at-Risk limits generated by the Autonomous Financial Risk Intelligence Engine comply with Basel Committee on Banking Supervision (BCBS) standards. The model has passed the Kupiec POF Likelihood Ratio test at a 95% confidence level.